

Mohammadreza Rezaei

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Research interests

- Machine Learning
- Deep Learning
- Graph Neural Network (GNN)
- Data Analysis

Education

Amirkabir University of Technology - Tehran Polytechnic

Bachelor of Science in Computer Engineering, GPA: 3.35/4 (Overall),

GPA of last four semesters: 3.82/4

Sep. 2019 – May. 2025

Research Experience

Undergraduate Research Assistant

Data Science Research Group | Supervisor: [Dr. Mostafa Haghiri Chehreghani](#)

Feb. 2024 – Apr. 2025

- Studying the properties of Graph Neural Networks (GNNs) and exploring their applications across various domains.
- Focusing on Comparing Graph Neural Networks (GNNs) with and without learnable weight matrices across various tasks. | [Github](#)

Projects

Humanoid robot following | [Github](#)

May. 2023

- Implemented a humanoid robot-following system using ROS, incorporating YOLOv8 for human detection, a PID controller for maze navigation, and a proportional controller for real-time human following.

Fuzzy Logic Self-Driving Car | [Github](#)

Apr. 2023

- Implemented a fuzzy logic-controlled self-driving car that can navigate a predetermined route without colliding with obstacles.

Gumapa game solution using Genetic algorithm | [Github](#)

Mar. 2023

- Implemented a Gumapa game solution utilizing a genetic algorithm, incorporating mutation, selection, and crossover to optimize strategies.

Pacman Search Algorithms | [Github](#)

Dec. 2022

- Implemented DFS, BFS, UCS, and A* with custom heuristics in a Pacman maze to study search optimality, completeness, and performance on problems such as corner navigation and food collection.

Plants vs Zombies game | [Github](#)

Nov. 2020

- Implemented the popular "Plants vs Zombies" game.

Relevant Courseworks

- Artificial Intelligence | A
- Robotics | A
- Principles of Computational Intelligence | A
- Image Processing & Neural Networks | A
- Data Mining | A
- Data Structures & Algorithms | A
- Algorithm Design | A
- Linear Optimization | A
- Stanford CS224W: Graph Neural Networks
- Deep Learning Specialization – Coursera

Honors & Awards

- Ranked in the **top 0.05%** in the Iranian University Entrance Exam, also known as Konkour.
- Founded and administered by National Organization for Development of Exceptional Talent. (NODET)

Teaching Experience

Teaching Assistant

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|---|-----------------------|
| · Principles & Applications of Artificial Intelligence Instructor: Dr. Mahdi Javanmardi | Feb. 2024 – Jul. 2024 |
| · Robotics Instructor: Dr. Mahdi Javanmardi | Sep. 2023 – Jan. 2024 |
| · Principles of Computational Intelligence Instructor: Dr. Mohammad Mehdi Ebadzadeh | Sep. 2023 – Jan. 2024 |
| · Software Engineering Instructor: Dr. Ehsan Alirezaei | Feb. 2023 – Jul. 2023 |

Technical Skills

Programming Languages: Python, Java, C++, C, R, Javascript, Rust, SQL, HTML5/CSS3

Tools & Technologies: Linux, Git & Github, $\text{\LaTeX}$, Pycharm, Jupyter, Anaconda

ML/DL Frameworks & Libraries: Pytorch, PyG, Tensorflow, Numpy, Pandas, OpenCV, Matplotlib, Seaborn, Scipy, Scikit-Learn

Languages

English: TOEFL(IBT): 104 (R:28 L:24 S:27 W:25)

Persian: Mother tongue

Volunteering works

- Technical Team Member for GameCraft 2022, Computer Engineering Department, Amirkabir University
- Executive Team Member for Amirkabir Artificial Intelligence Competition (AAIC)